

Archimedes: An Economical Photonic Demonstration of the Hadamard Gate in Linear Optical Quantum Computing (LOQC) Inspired by the KLM Protocol

Author: James Eckhart, Jr.

Independent Quantum Researcher
Senior Solution Engineer, Microsoft Corporation
March 2026

github.com/JamesEckhartJr | James@JamesEckhartJr.com

“The shortest path between two truths, in the real domain, passes through the complex domain -- Le chemin le plus court entre deux vérités, dans le domaine réel, passe par le domaine complexe.” - Jacques Hadamard (Paris, 1965)

Abstract

Linear Optical Quantum Computing (LOQC) offers a scalable, room-temperature route to quantum information processing by encoding qubits in photonic states and manipulating them with passive linear optics. This paper presents *Archimedes*, a low-cost, benchtop photonic demonstrator that implements a single-qubit Hadamard gate using dual-rail path encoding and principles inspired by the Knill–Laflamme–Milburn (KLM) protocol.

The system employs a pulsed 650 nm laser diode as a pseudo-single-photon source, together with 50/50 beam splitters, phase shifters, and an Arduino-based timing and detection architecture. While the laser produces coherent states rather than true single-photon Fock states, the experiment successfully demonstrates interference patterns consistent with the creation and measurement of photonic superposition states. Post-selection is used to emulate measurement-induced nonlinearity.

The project highlights both the accessibility of photonic quantum experiments and the practical challenges involved, including photon loss, mode mismatch, detector efficiency, and the need for high-quality single-photon sources in future iterations. *Archimedes* serves as an open-source educational platform for students and researchers exploring LOQC fundamentals.

Keywords: Linear Optical Quantum Computing; Photonic Quantum Computing; KLM Protocol; Hadamard Gate; Dual-Rail Encoding; Open-Source Quantum Hardware

1. Introduction

The Hadamard gate, named after the French mathematician *Jacques Hadamard*, is a fundamental single-qubit operation in quantum computing that transforms computational basis states into equal superpositions. In matrix form, the Hadamard operation is defined as:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Applied to the computational basis states, it yields:

$$H | 0 \rangle = \frac{| 0 \rangle + | 1 \rangle}{\sqrt{2}}, H | 1 \rangle = \frac{| 0 \rangle - | 1 \rangle}{\sqrt{2}}$$

In photonic quantum systems, the Hadamard gate can be implemented naturally using a 50/50 beam splitter when information is encoded in either polarization or spatial (dual-rail) modes. Linear Optical Quantum Computing (LOQC) leverages such passive optical elements—beam splitters, phase shifters, and mirrors—combined with measurement and post-selection to perform quantum logic.

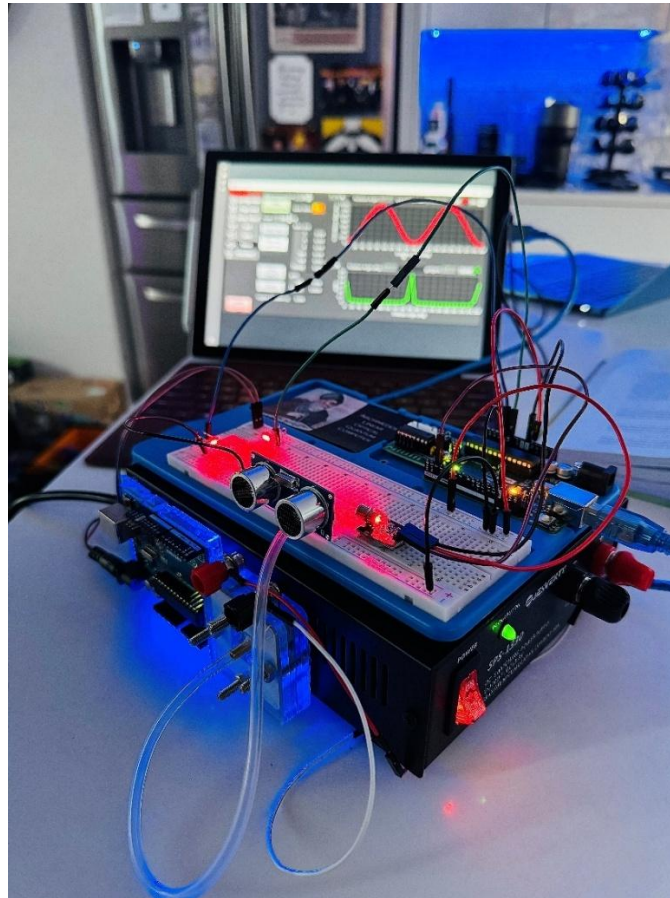


Figure 1. Archimedes Photonic Linear Optical Quantum Computer

The Knill–Laflamme–Milburn (KLM) protocol demonstrated that scalable quantum computation is theoretically possible using only linear optics, ancillary photons, projective measurements, and classical feed-forward control. In the KLM framework, single-qubit gates like the Hadamard are straightforward with linear optics, while two-qubit gates (e.g., controlled-phase or CZ gates) are probabilistic and rely on measurement-induced nonlinearity. For a CZ gate in KLM, the operation can be represented as a parity check using beam splitters and ancillary modes, with success probability scaling with the number of ancillae. A simplified KLM CZ gate involves interfering target and control photons with ancilla photons on beam splitters, followed by post-selection on detection outcomes. The effective nonlinearity arises from conditioning on specific measurement results, such as:

$$|\psi\rangle = \alpha |00\rangle + \beta |01\rangle + \gamma |10\rangle + \delta |11\rangle \rightarrow \text{post-selected on even parity detections}$$

While full KLM implementations require high-efficiency single-photon sources and fast optical switching, simplified demonstrations remain valuable for education, prototyping, and conceptual validation. This work introduces *Archimedes*, an economical and open-source photonic experiment designed to demonstrate a single-qubit Hadamard operation within the LOQC framework using readily available components. The system is intended as an accessible platform for students, researchers, and engineers exploring photonic quantum computing principles.

2. System Design

2.1 Photon Source and Encoding

The *Archimedes* system uses a pulsed 650 nm laser diode as a pseudo-single-photon source. Pulse timing is controlled via an Arduino microcontroller to produce short, repeatable optical pulses suitable for interference measurements.

Quantum information is encoded using dual-rail path encoding, where logical states are represented by spatial modes:

$$|0\rangle = |1\rangle_{\text{upper}} |0\rangle_{\text{lower}}, |1\rangle = |0\rangle_{\text{upper}} |1\rangle_{\text{lower}}$$

This encoding allows straightforward implementation of single-qubit gates using beam splitters and mirrors.

2.2 Optical Layout

A 50/50 beam splitter serves as the core optical element implementing the Hadamard transformation. When a photon enters one input port, the beam splitter produces an equal superposition of output paths, corresponding to the Hadamard operation on the dual-rail qubit:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Photon detection is performed using photodiode sensor modules positioned at the output paths. While true coincidence counting is not implemented, detection events are time-correlated and post-selected to emulate simplified KLM-style measurement behavior.

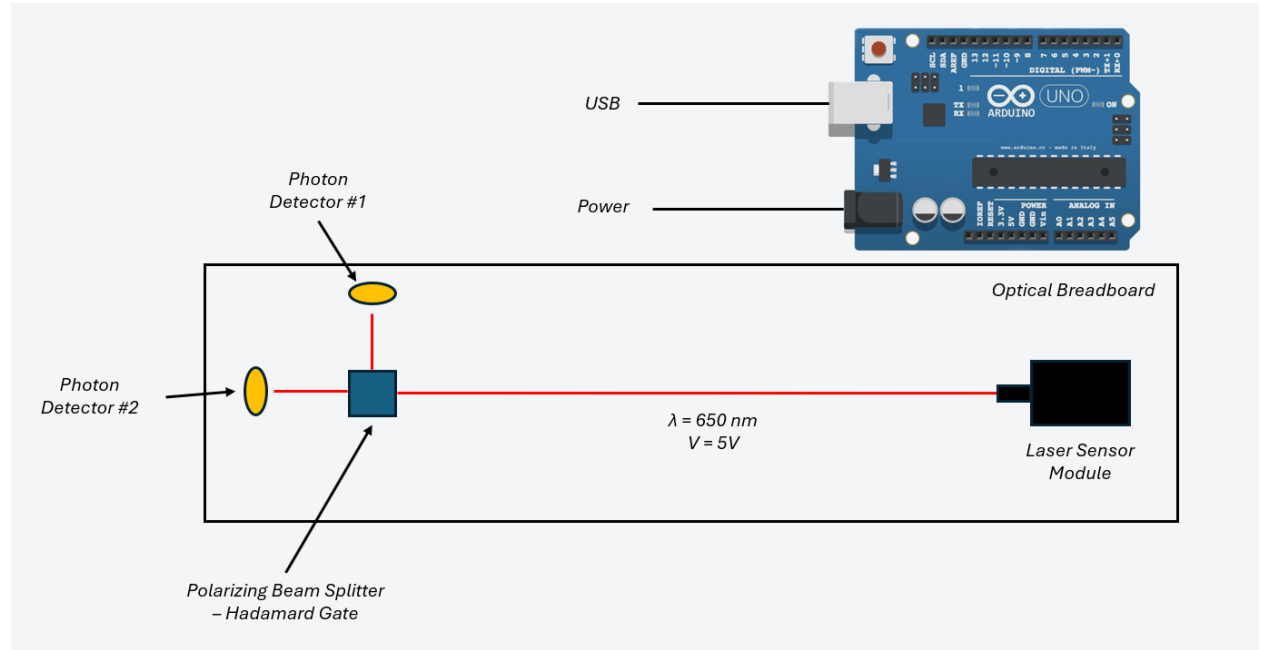


Figure 2. Optical layout of the Archimedes photonic Hadamard gate demonstration.

3. Experimental Implementation

The experimental procedure consists of four primary stages:

1. Input State Preparation: A coherent optical pulse is injected into a single rail, representing a computational basis state (e.g., $|0\rangle$).
2. Hadamard Gate Application: The beam splitter transforms the input state into a superposition across the two spatial modes:

$$|0\rangle \rightarrow \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

3. Post-Selection: Detection events consistent with expected interference behavior are selected to approximate KLM-style conditional operation.
4. Verification and Measurement: Path population statistics are recorded and analyzed to verify superposition behavior.

An Arduino microcontroller coordinates laser pulsing, sensor acquisition, and serial output. The embedded firmware pulses the laser, reads analog signals from the two output-path photodiodes, compares their relative intensities, and outputs the corresponding logical qubit state via the serial monitor.

```

1  /*
2  | * 'Archimedes' Linear Optical Photonic Processor
3  | * Author: James Eckhart, Jr.
4  | * "Rise above oneself and grasp the world." - Archimedes
5  | */
6  |
7  | const int TRIGGER_PIN = 2;    // Pulses the quantum circuit
8  | const int H_PIN = A0;        // Measures horizontally polarized photons
9  | const int V_PIN = A1;        // Measures vertically polarized photons
10 | const int LED_PIN = 13;      // Onboard LED for visual feedback
11 |
12 | void setup() {
13 |   pinMode(TRIGGER_PIN, OUTPUT);
14 |   pinMode(LED_PIN, OUTPUT);
15 |   Serial.begin(9600);
16 |
17 |   Serial.println("=== Archimedes Photonic Processor Initialized === FIRE THE LASER!!! -Dr. Evil");
18 | }
19 |
20 | int generateRandomBit() {
21 |   // Pulse the laser
22 |   digitalWrite(TRIGGER_PIN, HIGH);
23 |   delay(100);
24 |   digitalWrite(TRIGGER_PIN, LOW);
25 |
26 |   // Read photon polarization levels
27 |   float hValue = analogRead(H_PIN);
28 |   float vValue = analogRead(V_PIN);
29 |
30 |   // Determine random bit
31 |   if (hValue > vValue) {
32 |     digitalWrite(LED_PIN, LOW); // Indicate bit 0
33 |     return 0;
34 |   } else if (hValue < vValue) {
35 |     digitalWrite(LED_PIN, HIGH); // Indicate bit 1
36 |     return 1;
37 |   } else {
38 |     // Equal photon counts - retry until a bit is generated
39 |     return generateRandomBit();
40 |   }
41 | }
42 |
43 | void loop() {
44 |   int bit = generateRandomBit();
45 |   Serial.print(" Quantum Bit -- Ket: |");
46 |   Serial.print(bit);
47 |   Serial.println(">");
48 |   delay(1000); // Slight delay for readability
49 | }

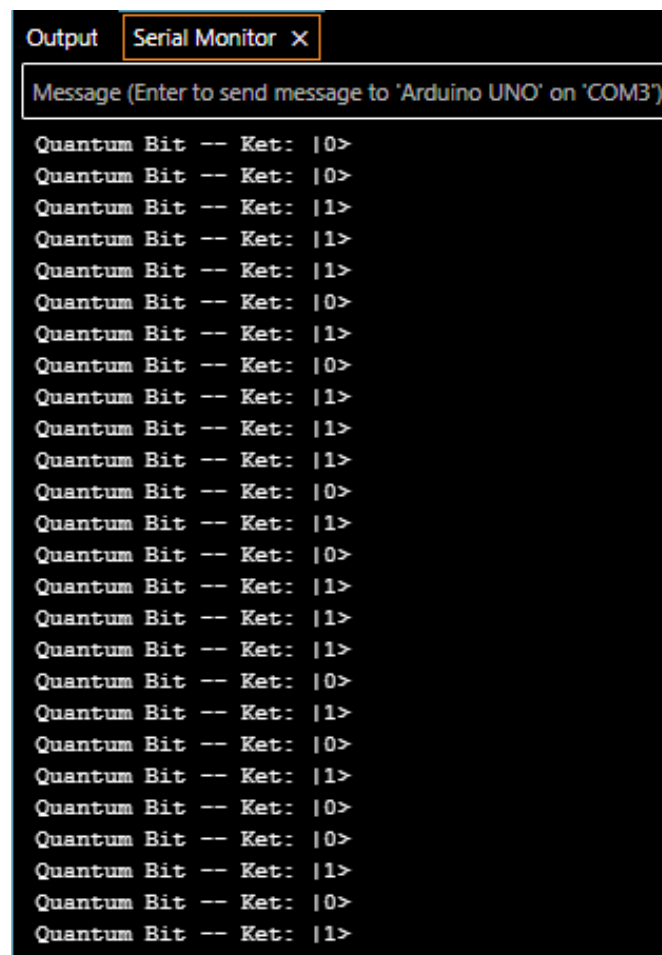
```

Figure 3. Arduino Microcontroller Code (C++)

4. Results and Discussion

The experiment demonstrates clear interference behavior consistent with Hadamard gate operation in a photonic system, with expected balanced detection events between the two logical states indicating successful generation of superposition states. The project highlights several practical challenges inherent to low-cost LOQC implementations, including optical loss, mode mismatch, detector inefficiency, and the absence of true single-photon sources. Nevertheless, the system effectively illustrates the core principles of photonic quantum logic and measurement-based operation.

While *Archimedes* does not implement a full KLM two-qubit gate, it provides a tangible foundation upon which more advanced photonic circuits—such as controlled-phase gates and entanglement sources—may be constructed.

A screenshot of a serial monitor window titled "Output Serial Monitor X". Below the title bar is a text input field with the placeholder "Message (Enter to send message to 'Arduino UNO' on 'COM3')". The main area of the window displays a list of 25 lines of text, each representing a quantum bit measurement. Each line consists of the text "Quantum Bit -- Ket:" followed by either $|0\rangle$ or $|1\rangle$. The sequence of states is: $|0\rangle, |0\rangle, |1\rangle, |1\rangle, |1\rangle, |0\rangle, |1\rangle, |0\rangle, |1\rangle, |1\rangle, |1\rangle, |0\rangle, |1\rangle, |0\rangle, |1\rangle, |1\rangle, |1\rangle, |0\rangle, |1\rangle, |0\rangle, |1\rangle, |0\rangle, |0\rangle, |1\rangle, |0\rangle, |1\rangle$.

```
Output Serial Monitor X
Message (Enter to send message to 'Arduino UNO' on 'COM3')
Quantum Bit -- Ket: |0>
Quantum Bit -- Ket: |0>
Quantum Bit -- Ket: |1>
Quantum Bit -- Ket: |1>
Quantum Bit -- Ket: |1>
Quantum Bit -- Ket: |0>
Quantum Bit -- Ket: |1>
Quantum Bit -- Ket: |0>
Quantum Bit -- Ket: |1>
Quantum Bit -- Ket: |1>
Quantum Bit -- Ket: |1>
Quantum Bit -- Ket: |0>
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Quantum Bit -- Ket: |0>
Quantum Bit -- Ket: |1>
Quantum Bit -- Ket: |0>
Quantum Bit -- Ket: |0>
Quantum Bit -- Ket: |1>
Quantum Bit -- Ket: |0>
Quantum Bit -- Ket: |1>
```

Figure 4. Quantum Bit Output from Serial Monitor -- the states $|0\rangle$ (ket zero) and $|1\rangle$ (ket one) represent the logical basis states of a qubit encoded in the dual-rail (path) encoding scheme.

5. Conclusion

Archimedes provides a tangible, replicable demonstration of photonic quantum logic, inspired by the KLM framework. The project validates the feasibility of benchtop photonic quantum experiments and serves as an educational and prototyping platform for exploring linear optical quantum computing fundamentals.

All design files, firmware, schematics, Jupyter analysis notebooks, and supporting materials are released as open-source resources available at <<https://github.com/JamesEckhartJr>> to encourage replication, extension, and further experimentation in photonic quantum computing.

References

1. Knill, E., Laflamme, R., & Milburn, G. J. (2001). A scheme for efficient quantum computation with linear optics. *Nature*, 409, 46–52.
2. Additional references and implementation details are available in the project repository: <<https://github.com/JamesEckhartJr>>.